YEAR 7 · MATHS UNIT 5 · SCIENCE UNIT 3 · HOMEWORK 11

180° Apart

Part 1: Maths — 5.1 A sum of 360° and 5.2 Intersecting lines
Part 2: Science — 3.4 Tides, 3.5 Energy and 3.6 Changes in energy
Part 3: Maths and science together

Name: _____ **Class:** _____

Date given: _____ **Date due:** Friday 6 November 2026

How to do this homework

- **A–B:** maths. **C–E:** science. **F:** both together. **No calculator.** Write the **calculation** first, then the answer.
- **Give a reason** = write the angle fact after **because**: *"a = 40° because they are vertically opposite angles."*
- Answer in **full sentences** when it says so. *"Kinetic"* is not a sentence. *"The bike has kinetic energy because it is moving"* is.

Section A · Maths 5.1 — A Sum of 360°

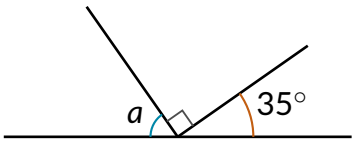
A1 Find the missing angle: two steps

Two steps, from the lesson

1. Add the angles you know. **2. Subtract** the total from **180°** (a straight line) or **360°** (around a point, or in a quadrilateral).
In a **parallelogram**, opposite angles are **equal**.

The diagrams are drawn accurately. Find each letter. (a) is done for you.

(a)

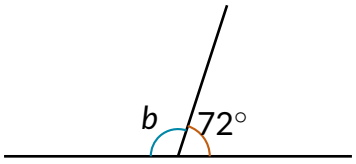


1. 35° + 90° = 125°

2. 180° − 125° = 55°

a = 55°

(b)

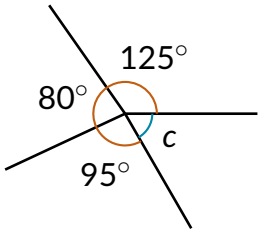


1. _____

2. _____

b = _____

(c)

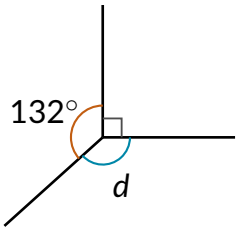


1. _____

2. _____

c = _____

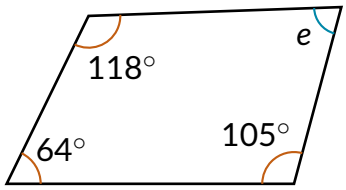
(d)



1. _____
2. _____

d = _____

(e)



1. _____
2. _____

e = _____

(f)



a parallelogram

q = _____

1. _____

p = **r** = _____

A2 From English to a calculation

Word help — the words in angle questions

sum	the total when you add	**work out**	find, and show the calculation
equal	the same size	**a full turn**	360°

Write the **calculation first**. Then write the answer.

(a) Two angles are 63° and 116°. Mr Bowen says they make a **straight line**. Is he right?

Calculation: _____ **Answer:** _____

(b) The sum of two angles is 180°. One of them is 64°. Work out the other one.

Calculation: _____ **Answer:** _____

(c) Three angles of a quadrilateral are 90°, 90° and 47°. Calculate the fourth angle.

Calculation: _____ **Answer:** _____

(d) One angle of a quadrilateral is 120°. The other three angles are equal. Work out the size of each one.

Calculation: _____ **Answer:** _____

(e) Mr Bowen spins round on his chair. He turns through 1080°. How many full turns is that?

Calculation: _____ **Answer:** _____

A3 Now you write the question

Write a word problem for $360° - 70° - 110° - 95° = 85°$

Your word problem must have: tick each one

- ☐ a real thing (a garden, a kite …)
- ☐ the word **quadrilateral**
- ☐ three angles
- ☐ full sentences
- ☐ a question at the end
- ☐ the answer 85°

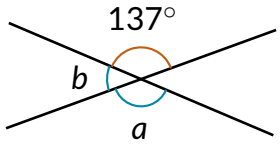
Section B · Maths 5.2 — Intersecting Lines

B1 Find the angle. Give a reason.

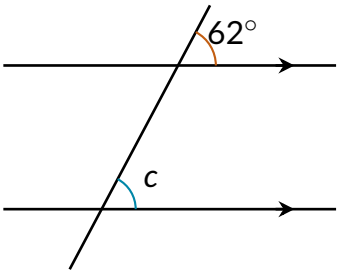
The four reasons — copy the one you need after because

X: vertically opposite angles are equal
F: corresponding angles are equal
line: angles on a straight line add up to 180°
Z: alternate angles are equal
Example: *a* = 137° because vertically opposite angles are equal.

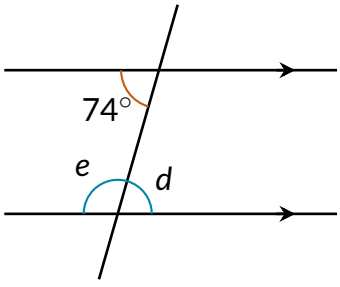
Arrows (>) on two lines mean they are **parallel**. Find each letter, and give a reason.



(i) two lines cross



(ii) parallel lines



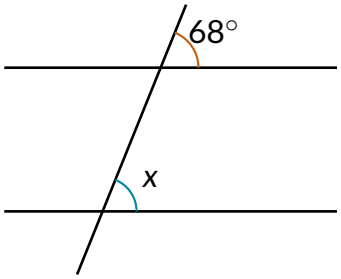
(iii) parallel lines

b = _____ because _____

c = _____ because _____

d = _____ because _____

e = _____ because _____



(iv) Which statement is correct: **A**, **B** or **C**?

A $x = 68°$

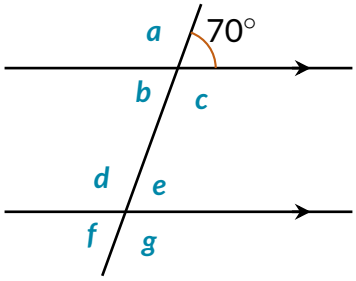
B $x = 112°$

C You need more information.

Answer: _____ because _____

_____

B2 Mr Bowen's homework



Only **one** of Mr Bowen's sentences is completely right. Circle **right** or **wrong**. If it is wrong, write the correct sentence.

(a) *b* = 70° *because they are corresponding angles.* right · wrong

_____

(b) *e* = 70° *because they are alternate angles.* right · wrong

_____

(c) *a* = 70° *because vertically opposite angles are equal.* right · wrong

_____

(d) *d* = 110° *because d and c are alternate angles.* right · wrong

_____

Section C · Science 3.4 — Tides

C1 A day at the harbour

Mr Bowen writes down the tides at his harbour for one day.

Time	3 a.m.	9 a.m.	3 p.m.	9 p.m.
Tide	high tide	low tide	high tide	low tide
Depth	5.7 m	1.3 m	5.7 m	1.3 m

(a) How many high tides are there in one day? _____

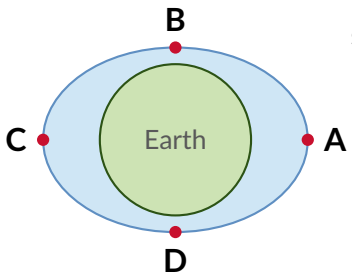
(b) How many hours is it from high tide to low tide? _____

(c) Work out the **tidal range**. (**tidal range**: the difference in depth between high tide and low tide)

Calculation: _____ **Answer:** _____

(d) Tomorrow, will the first high tide be at 3 a.m. again? Answer in a full sentence. Start: **"No, because the Moon …"**

C2 Two high tides at the same time



seen from above the North Pole

not to scale



(a) Is each place at **high tide** or **low tide**? Write **H** or **L**.

A ☐ **B** ☐ **C** ☐ **D** ☐

(b) The Moon is on the right. Why is there a high tide at **C**, on the far side? Start: **"The near side is pulled the most. The far side …"**

C3 The Sun, and the land

(a) The Sun's gravity is **stronger** than the Moon's. Why does the Moon make **bigger** tides? Start: **"The Sun is much further away, so …"**

(b) The land in Hanoi goes up and down about **30 cm**, twice a day. Why do we **not** feel it? Start: **"We do not feel it because …"**

Section D · Science 3.5 — Energy

D1 Mr Bowen's afternoon in joules

The joules, from the lesson

1 floor of stairs: **2 000 J** · run 1 metre: **200 J** · boil 1 litre of water: **400 000 J**

Write the **calculation first**. Then write the answer in joules (J).

(a) He climbs 4 floors to the science lab.

Calculation: _____ **Answer:** _____

(b) He runs 45 m to catch the bus. He misses it.

Calculation: _____ **Answer:** _____

(c) He boils 0.5 litres of water for his tea.

Calculation: _____ **Answer:** _____

(d) How many **times as much** energy does the tea use as the stairs? (**how many times as much**: divide)

Calculation: _____ **Answer:** _____

D2 Mr Bowen's bike ride at night

Name the energy in each clue. Then write **store** or **transfer**. Row 1 is done for you.

Clue	Energy	Store or transfer?
1. The bike is moving.	kinetic	store
2. Mr Bowen is at the top of a hill.		
3. The banana in his bag.		
4. The bike lamp shines.		
5. A current flows in the wire to the lamp.		
6. He rings the bell.		
7. He brakes. The brakes get hot.		
8. The tyres are squashed.		

D3 Which stores last?

Will it keep its energy for a **year**? Write **Y** (yes) or **N** (no).

a cup of hot tea ☐ a packet of rice ☐ a spinning top ☐ a rock at the top of a cliff ☐

Section E · Science 3.6 — Changes in Energy

E1 An event or a process?

An **event** happens once. A **process** keeps going. Write **E** (event) or **P** (process) in each box.

a plate falls and breaks ☐ a balloon pops ☐ a candle burns ☐

the Sun shines ☐ a firework explodes ☐ a fridge keeps food cold ☐

E2 Energy-change diagrams

Write the missing energy in each empty box. Read each arrow as **to**.

(a) A toaster



(b) Charging a phone



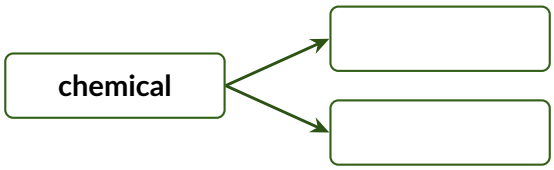
(c) A skateboard rolls down a ramp



(d) Mr Bowen rubs his hands



(e) A candle burns. Two arrows from one box.



E3 The bouncing football

Mr Bowen drops a football from **2 m**. It bounces up to **1.3 m**.

(a) Which store is the **biggest** at each moment? Use: **kinetic** · **elastic potential** · **gravitational potential**

Moment	Biggest store
in his hand, 2 m up	gravitational potential
just before it hits the floor	
squashed on the floor	
at the top of the bounce	

(b) How much lower than 2 m does it bounce?

Calculation: _____ **Answer:** _____

(c) Why does it not bounce back up to 2 m? Answer in a full sentence. Start: **"At the bounce, some energy …"**

E4 Hotter to colder

Mr Bowen's classroom is at 30 °C. Does thermal energy move **into** each drink, or **out of** it? Circle one.

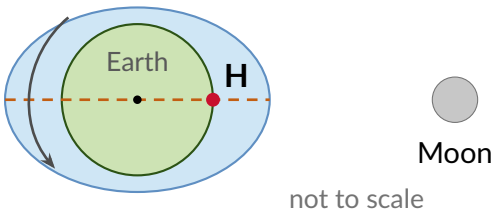
(a) a hot coffee at 80 °C **into** · **out of**

(b) a cold can of juice at 4 °C **into** · **out of**

(c) Mr Bowen opens the fridge. He says: *"Close the door! You are letting the cold out!"* Is he right? Explain. Start: **"He is …, because thermal energy …"**

Section F · Maths and Science Together

F1 180° apart



The Earth seen from above the North Pole. The two bulges are on a **straight line** through the middle of the Earth. Mr Bowen's harbour **H** has high tide now.

The Earth **spins** once a day: **360° in 24 hours**.

(a) How many degrees does the Earth turn in **1 hour**?

Calculation: _____ **Answer:** _____

(b) Complete the table.

Hours	1	3	6	12	24
Degrees	15°				360°

(c) After **6 hours**, H has turned through a **right angle**. Is H at high tide or low tide then?

_____

(d) After **12 hours**, H has turned 180°. It is on the other side of the Earth. Is H at high tide or low tide?

_____

(e) Complete: The two bulges are _____° apart. So H has a high tide every _____ hours.

(f) Every day the Moon moves on a little. So the Earth must turn about **12° more** to bring H back to the bulge. The Earth turns 1° every **4 minutes**. How many minutes later is high tide each day?

Calculation: _____ **Answer:** _____

F2 In a line, or at right angles?



(i) full moon **(ii)** half moon

seen from above the North Pole, not to scale

(a) Write the angle **?** at the Earth, between the line to the Sun and the line to the Moon.

(i) _____ **(ii)** _____

(b) In **(ii)**, the two lines are **parallel** · **perpendicular**. Circle one.

(c) In the lesson's model, the tidal range at Mr Bowen's harbour was **6 m** in one picture and **2 m** in the other. Which picture had the 6 m tides? _____

(d) How much **bigger** is the 6 m range? How many **times as big**?

Bigger by: _____ **Times as big:** _____

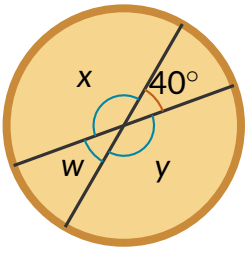
(e) Why are the tides biggest when the Sun, the Moon and the Earth are in a line? Start: **"When they are in a line, the Sun's tidal force …"**

F3 A slice of energy

Food has a **chemical** store. Mr Bowen's pizza stores **3 600 000 J**. So every **1°** of pizza stores **10 000 J**.

(a) He cuts it into **8 equal slices**. What angle is one slice? How many joules does one slice store?

Angle: _____ **Joules:** _____



(b) The next pizza: he makes **two straight cuts**. One slice is 40°. Find *w*, *x* and *y*. Give a reason for each.

w = _____ because _____

x = _____ because _____

y = _____ because _____

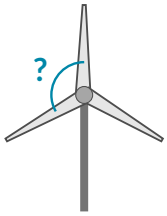
(c) He eats the **two biggest** slices. How many degrees of pizza is that? How many joules?

Degrees: _____ **Joules:** _____

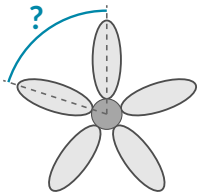
(d) One floor of stairs uses **2 000 J**. How many floors could he climb with the energy in those two slices? (Landmark 81 has 81 floors.)

Calculation: _____ **Answer:** _____

F4 The wind turbine and the fan



a **wind turbine**: 3 blades



a **ceiling fan**: 5 blades

(a) The blades are **equally** spaced. Work out each angle **?**

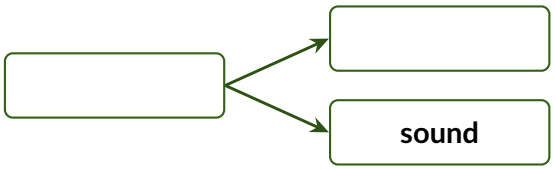
Turbine: _____ **Fan:** _____

(b) Fill in the energy-change diagrams. Use: **kinetic** · **electrical** · **sound**

Turbine



Fan



(c) What do you notice about the two diagrams? Answer in a full sentence. Start: **"The fan's energy change is …"**

F5 Explain it with maths

Mr Bowen says: *"The Moon is on one side of the Earth. So there is only **one** high tide a day."* Write him a reply in English: 4 or 5 sentences.

Your reply must have: tick each one

☐ **tidal force**	☐ **two bulges**	☐ **180°** apart
☐ the Earth **spins**	☐ **12 hours**	☐ full sentences

Answer Key — Teacher Copy

A1 Two steps

(b) 180° − 72° = **108°** (c) 125 + 80 + 95 = 300, 360 − 300 = **60°** (d) 90 + 132 = 222, 360 − 222 = **138°**
(e) 64 + 105 + 118 = 287, 360 − 287 = **73°** (f) q = **58°** (opposite); 360 − 58 − 58 = 244, $p = r$ = 244 ÷ 2 = **122°**

Every diagram is drawn accurately, so a protractor checks any answer — worth saying to a student who is stuck.
(b) has one known angle, so step 1 is just 72°; expect 252° (180 + 72) from anyone who adds when they should subtract.
(c) and (d) are *round a point*: expect 180 − 300 (impossible) or 180 − 222 from a student who uses 180 for everything. The fix is the first question of the two steps: *180 or 360?*
(d) has a small square: a right angle, 90°, not written as a number. Expect 228° (360 − 132) from anyone who skipped it.
(f) is the deck's parallelogram with new numbers; p = 122 and q = 122 means the student paired the wrong corners (opposite = across the shape, not next to it).

A2 From English to a calculation

(a) 63 + 116 = 179°: **no**, a straight line is 180° (b) 180 − 64 = 116°
(c) 360 − (90 + 90 + 47) = 360 − 227 = 133° (d) 360 − 120 = 240, 240 ÷ 3 = 80° (e) 1080 ÷ 360 = 3 full turns

Mark the calculation, not the answer.
(a) is the deck's vote (57 + 121 = 178) with new numbers. It looks straight; the point is that you *add*, you do not look. "Yes" with no calculation is the error the vote was for.
(b) "The sum of two angles is 180" is a straight line without the picture. A student who writes 180 + 64 read *sum* and added the numbers they could see: have them read the word-help line aloud.
(d) is two steps and the second is a division: expect 240°, which is the three angles together.
(e) runs the deck's chair backwards (there: 2.5 × 360 = 900). Expect 1080 × 360 from a student who sees two numbers and multiplies. He is facing the board again; the lesson has not moved.

A3 Now you write the question

Check, in order: a quadrilateral (or a four-sided real thing called one); three angles 70°, 110°, 95°; it asks for the fourth; it ends in a question mark; it uses full sentences. 70 + 110 + 95 + 85 = 360.
Model: *"Mr Bowen's garden is a quadrilateral. Three of its angles are 70°, 110° and 95°. What size is the fourth angle?"*
The usual failure is a question whose answer is not 85° — e.g. one that gives all four angles, or asks for their sum. A question about a triangle has the wrong total; ask the student what 360° is for.

B1 Find the angle. Give a reason.

b = 43° because angles on a straight line add up to 180° (180 − 137).
c = 62° because corresponding angles are equal.
d = 74° because alternate angles are equal.
e = 106° because angles on a straight line add up to 180° (180 − 74).
(iv) **C**, because there are no arrows, so we do not know that the lines are parallel.

Accept the reason in the student's own words if the fact is right ("they make a Z" is fine for d). A right number with no reason is half done: the reason is the 5.2 skill.
b = 137° means the student called every angle at an X "vertically opposite". b is *next to* 137°, not

opposite it.
(iv) The lines look parallel and 68° is tempting; **A** is the error the deck's slide 21 was for. **B** (112°) is worse: it is a guess with a subtraction in it.

B2 Mr Bowen's homework

(a) **wrong**: b = 70° because *vertically opposite* angles are equal.
(b) **wrong**: e = 70° because they are *corresponding* angles.
(c) **wrong**: a = 110° because angles on a straight line add up to 180°.
(d) **right** (c = 110°, and c and d make a Z).

(a) and (b) have the right *number* and the wrong *name*. A student who writes "right" for them is checking the arithmetic, not the reason: point at the word "completely".
(c) is the trap from the other side: a is *next to* the 70°, so it is 110°. A correction that keeps 70° and changes only the reason has missed it.
(d) needs c first, and c's size is not given. Students who mark it wrong because "we don't know c" should find it (180 − 70) and look again.

C1 A day at the harbour

(a) 2 (b) 6 hours (c) 5.7 − 1.3 = 4.4 m
(d) *No, because the Moon moves on a little every day, so high tide is about 50 minutes later (about 3:50 a.m.).*

(b) Expect 12 (high to *high*).
(d) Accept any answer that says "later" and gives the Moon moving as the reason; the 50 minutes is a bonus. "The Earth moves" alone is not the reason — it moves every day.
The book's Question 6 (average depth and range: not 7.0 m) is Extra Homework 11's X3, so it is not repeated here.

C2 Two high tides at the same time

(a) A **H** · B **L** · C **H** · D **L**
(b) *The near side is pulled the most. The far side is pulled the least, so it is left behind, and the water makes a bulge there too.*

(a) C written as L is the deck's vote all over again ("one bulge"). If it is common, act out the three students again: 3 steps, 2 steps, 1 step. The middle is the Earth; the 1-step student is the far bulge, left behind. Section F1 and F5 lean on this.

C3 The Sun, and the land

(a) *The Sun is much further away (390 times), so it stretches the Earth less.*
(b) *We do not feel it because everything goes up and down together: the ground, the buildings and us.*

(a) "The Sun is far away, so its gravity is weak" contradicts the question (it says the Sun's pull is stronger). The word to look for is *stretch*: tides come from the stretch, not the pull.

D1 Mr Bowen's afternoon in joules

(a) 4 × 2 000 = 8 000 J (b) 45 × 200 = 9 000 J (c) 0.5 × 400 000 = 200 000 J
(d) 200 000 ÷ 8 000 = 25 times

(c) is half of 400 000 (4.5's quick way: ×0.5 is half). Expect 2 000 000 or 20 000 from a student who moved the point instead of halving.
(d) is "how many times": divide. 200 000 − 8 000 = 192 000 is the HW 10 trap (*how much more* against *how many times as much*) again.

D2 Mr Bowen's bike ride at night

2 gravitational potential, store · 3 chemical, store · 4 light, transfer · 5 electrical, transfer · 6 sound, transfer · 7 thermal, store (accept transfer) · 8 elastic potential, store

Row 3 is food, and food is chemical: "kinetic" (because Mr Bowen is riding) has read the wrong part of the clue. Row 7 is thermal either way: the hot brakes store it, and they heat the air. Accept both.

D3 Which stores last?

tea **N** · rice **Y** · spinning top **N** · rock **Y**

The write note: chemical and gravitational potential last; thermal and kinetic do not. A student who writes Y for the tea is thinking of the drink, not its heat: the drink lasts, its thermal store goes into the room in an hour (E4 asks which way).

E1 An event or a process?

plate **event** · Sun **process** · balloon **event** · candle **process** · firework **event** · fridge **process**

E2 Energy-change diagrams

(a) thermal · (b) chemical · (c) gravitational potential · (d) kinetic → thermal · (e) light and thermal (either order)

(b) is the deck's vote: *electrical* in the empty box means the student thinks a battery stores electricity. It stores *chemical* energy.
(a) accept "thermal" or "thermal and light" (a glowing toaster).
(d) is the deck's "rub your hands": kinetic *to* thermal.

E3 The bouncing football

(a) kinetic · elastic potential · gravitational potential
(b) 2 − 1.3 = 0.7 m
(c) *At the bounce, some energy changed to sound and heat, so it cannot lift the ball back up to 2 m.*

(b) Expect 1.1 (2 − 1.3 worked as 3 − 2, or 13 − 2) from a student who does not write 2 = 2.0.
(c) "Gravity pulls it down" is not the reason — gravity pulled it down on the way *down* too. "The energy is lost" is half: say where it went.

E4 Hotter to colder

(a) out of · (b) into
(c) *He is wrong, because thermal energy moves from hotter things to colder things. Heat from the warm room goes into the fridge; cold does not come out.*

(c) is the freezer vote in 3.6 ("does cold go in, or heat come out?"). "He is right because the food gets warm" gets the result right and the direction wrong.

F1 180° apart

(a) 360 ÷ 24 = 15° (b) 3 h 45° · 6 h 90° · 12 h 180°
(c) low tide (d) high tide (e) 180°; every 12 hours
(f) 12 × 4 = 48 minutes: *about 50 minutes later*, like the lesson.

This is the question the packet is named for. The two bulges are on one straight line through the Earth, so they are 180° apart, and the Earth needs 12 hours to turn 180°. That is why high tide comes every 12 hours and low tide 6 hours (a right angle) after it.
(d) "Low" means the student thinks of one bulge (C2 again).
(f) The 12 is 360 ÷ 29: the Moon goes round in about 29 days. The 4 minutes is 24 × 60 = 1 440 minutes ÷ 360. Both are worth asking the strongest students to check. 48 is the deck's "about 50" (24 hours ÷ 29).

F2 In a line, or at right angles?

(a) (i) 180° · (ii) 90° (b) perpendicular
(c) (i) (d) 6 − 2 = 4 m bigger; 6 ÷ 2 = 3 times as big
(e) *When they are in a line, the Sun's tidal force pulls the oceans the same way as the Moon's, so they add up and the bulges are bigger.*

The 6 m and 2 m are the deck's *Line Them Up* model (Mr Bowen's harbour). New moon is in a line too: the angle at the Earth is 0° there, which is worth raising with anyone who asks.
(d) is the HW 10 pair again: *how much bigger* subtracts, *how many times* divides.

F3 A slice of energy

(a) 360 ÷ 8 = 45°; 45 × 10 000 = 450 000 J
(b) w = 40° because vertically opposite angles are equal; x = 140° because angles on a straight line add up to 180°; y = 140° (vertically opposite x, or a straight line)
(c) 140 + 140 = 280°; 280 × 10 000 = 2 800 000 J
(d) 2 800 000 ÷ 2 000 = 1 400 floors

(a) One slice (450 000 J) is more than boiling a litre of water (400 000 J): worth saying aloud.
(b) is the 5.2 deck's pizza with new numbers.
(d) is 3.5's kettle-to-stairs comparison; 1 400 floors is 17 Landmark 81s. A real body turns most of that energy into heat, not stairs — 3.7 will say so. He takes the lift.
Expect 28 000 000 or 280 000 in (c) from a student who loses count of zeros: ×10 000 is four zeros.

F4 The wind turbine and the fan

(a) turbine 360 ÷ 3 = 120° · fan 360 ÷ 5 = 72°
(b) turbine: kinetic → electrical · fan: electrical → kinetic (and sound)
(c) *The fan's energy change is the turbine's backwards: the turbine changes kinetic energy to electrical energy, and the fan changes electrical energy to kinetic energy.*

The turbine is the 3.6 deck's Bac Lieu wind farm. A student who draws "wind" in the turbine's first box has the idea; ask what kind of energy moving air has.

F5 Explain it with maths

Check the six ticks. A model reply: *"There are two high tides a day, not one. The Moon's tidal force pulls the oceans into two bulges, one on each side of the Earth. The bulges are on a straight line, 180° apart. The Earth spins 15° every hour, so it takes 12 hours to turn 180° from one bulge to the other. So every harbour has a high tide every 12 hours."*
The thinking to look for is the *why* of the far bulge (the stretch, from C2) joined to the *when* (12 hours, from F1). A reply that only says "the Earth spins" explains one high tide a day, which is Mr Bowen's mistake.